

TEDROS TESFU

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PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with 8+ years building production-grade microservices, event-driven architectures, and AI-powered applications. Expertise in Node.js/TypeScript backend development, Java Spring Boot, Python Flask, AWS serverless infrastructure, and LLM integration (Claude, OpenAI, prompt engineering). Delivered fault-tolerant systems processing financial transactions, led Angular-to-React migrations, and built AI-driven fitness and document automation platforms. Strong advocate for clean architecture, automated testing, and operational excellence.

TECHNICAL SKILLS

Languages & Frameworks: Node.js, TypeScript, JavaScript, Java (Spring Boot/Framework), Python (Flask, FastAPI), React, Redux, Angular, HTML5, CSS3, SASS

AI & LLM Integration: Claude API, OpenAI (GPT-4, GPT-3.5), Grok, Prompt Engineering, RAG Systems, Vector Embeddings, LLM Application Development

Cloud & Infrastructure: AWS (Lambda, S3, Step Functions, EventBridge, SNS, SQS, CloudWatch), Kafka, Redis, WebSockets, Socket.io

Architecture & APIs: Microservices, Event-Driven Architecture, REST, GraphQL, gRPC, SOLID, DRY, Domain-Driven Design

Data & Persistence: PostgreSQL, MongoDB, MySQL, Supabase, Hibernate, JPA

DevOps & Tools: Jenkins, GitHub Actions, GitLab, Docker, Kubernetes, Terraform, Jest, Mockito, Mocha, Jira, Rally, Confluence, Cursor, Claude Code, Codex, Blender (3D Modeling)

PROFESSIONAL EXPERIENCE

Software Engineer — John Deere Financial Co. | 2021 – 2024

- Ensured high availability for docket event processing by deploying event-driven microservices (Kafka, Node.js, TypeScript) across 3 AWS availability zones, automating file generation and SNS notifications, reducing manual overhead significantly
- Optimized delinquency management performance by pre-fetching and staging account data in S3 ahead of scheduled job execution, eliminating API bottlenecks and rate-limiting during peak processing windows
- Enforced data correctness through schema validation and business rule checks in Lambda functions, preventing malformed records from downstream propagation and protecting financial data integrity
- Implemented S3 retry buckets for failed account batches to ensure durability, enabling automatic reprocessing, comprehensive audit trails, and compliance documentation without manual intervention
- Reduced infrastructure costs by migrating from 24/7 on-premise servers to serverless event-driven Lambda, lowering spend while improving scalability via pay-per-execution pricing

- Built real-time letter generation microservice implementing business rule validation, XML payload generation, and secure SFTP transfers with comprehensive S3 backup archival for compliance and audit requirements
- Developed Java Spring Boot microservices for delinquent account processing and Debt Manager charge-off workflows, ensuring reliable handling of high-volume financial data with robust error handling and monitoring
- Created proof-of-claim processing service using Spring Boot, leveraging internal APIs to populate account data and storing records in S3 for downstream litigation team workflows
- Built AWS Lambda REST APIs for payload validation with CloudWatch monitoring and exponential backoff retry logic, ensuring fast, secure, and fault-tolerant validation during high-demand periods
- Developed AWS Step Functions orchestration workflows scheduled via EventBridge to automate critical tasks including S3→SFTP letter transfers, delinquent account loading, and proof of claim submissions, eliminating significant manual operations
- Established Jenkins/GitHub CI/CD pipelines with automated testing and blue-green deployments, dramatically reducing deployment time and enabling more frequent, reliable releases

Senior Software Developer (Contract) — Telostouch Inc. Toronto, ON | March 2021 – September 2021

- Led migration of large-scale Angular application to React, substantially reducing bundle size and improving page load performance and developer velocity
- Designed and delivered customer-facing features including user timelines, registration flows, dashboards, and profile management, serving thousands of daily active users
- Integrated Keycloak for centralized authentication and authorization (RBAC, SSO), consolidating identity management across multiple microservices and significantly reducing security vulnerabilities

Full-Stack Engineer (Contract) — Passionfruit Inc. Toronto, ON | September 2020 – March 2021

- Refactored real-time chat architecture from Agora to Socket.io/Node.js stack, substantially reducing message latency and supporting thousands of concurrent connections
- Implemented horizontal scaling strategy with Redis pub/sub for real-time messaging, enabling significant system growth without performance degradation

Full-Stack Engineer — Ministry of Education Asmara, Eritrea | 2017 – 2018

- Developed REST APIs and database integrations for educational gaming platform serving thousands of students, substantially improving content delivery performance
- Built comprehensive bug tracking and feedback system connecting hundreds of teachers with development team, dramatically reducing bug resolution time

Java Developer — Eritrean Trading Corporation Asmara, Eritrea | 2015 – 2016

- Designed and implemented Logistics Management System for warehouse operations, substantially reducing inventory tracking errors and improving order fulfillment efficiency
- Created comprehensive technical documentation and onboarding materials, enabling junior developers to independently maintain and extend system functionality

EDUCATION

Bachelor of Science in Computer Engineering | Eritrean Institute of Technology | 2015